

$$\int_0^1 \sqrt{1-\sqrt[3]{1-x}}dx$$

$$u:=1-x$$

$$du=-dx$$

$$\int_0^1 \sqrt{1-\sqrt[3]{u}}du$$

$$y=\sqrt{1-\sqrt[3]{u}}$$

$$y^2=1-\sqrt[3]{u}\forall y\text{ s.t. }1-\sqrt[3]{u}\geq 0$$

$$\sqrt[3]{u}\leq 1$$

$$u\leq 1\in [1,0]$$

$$u=(1-y^2)^3$$

$$du=-6y(1-y^2)^2dy$$

$$du=-6y(1-2y^2+y^4)dy$$

$$du=-6(y-2y^3+y^5)dy$$

$$-6\int_1^0 y(y-2y^3+y^5)dy$$

$$6\int_0^1 (y^6-2y^4+y^2)dy$$

$$6(\frac{1}{7}y^7-\frac{1}{5}y^5+\frac{1}{3}y^3)|_0^1$$

$$\frac{6}{7}-\frac{6}{5}+\frac{1}{3}$$

$$\frac{16}{35}$$